

Pooja Podduturu

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Education

University of North Texas, Denton

Texas, USA

Master in Health Data Analytics, CGPA : 4.0

Government Dental College and Hospital, Hyderabad

Aug 2024 - present

Bachelor's in dental surgery

Hyderabad, India

Aug 2017 - Dec 2023

EXPERIENCE

Research Assistant

Feb 2025 – May 2025

Department of Rehabilitation and Health Services, University of North Texas

Texas, USA

- Assisted in developing, revising, and updating a primary healthcare data collection questionnaire (60+ items), ensuring compliance with NIH and HIPAA standards, which improved response validity by 15% during pilot testing.
- Conducted statistical analysis of healthcare access disparities using SAS/SPSS, analyzing 5,000+ cases from national datasets. Identified socioeconomic and geographic predictors explaining 60% of variance in healthcare access across U.S. populations.
- Contributed to IRB documentation and supported the development of a manuscript focused on health behaviors and conditions prevalent among college students experiencing housing instability.
- Assisted Dr. Iwundu in drafting two systematic review papers:
 - Prevalence of Cataracts among Homeless populations: a systematic review — screened 1,200+ abstracts, extracted data from 35 studies, and synthesized prevalence and access-to-care outcomes.
 - NHANES-based analysis of cataract prevalence among U.S. women — analyzed 7,500+ participants, identifying sociodemographic and clinical factors associated with cataract prevalence.

Junior Dentist

Jan 2024 – May 2024

Vibrant Dental Clinic

Hyderabad, India

- Recorded and examined patient treatment data to track trends in dental conditions and treatment outcomes. Managed and updated Electronic Health Records (EHR), ensuring accurate data entry and compliance with health informatics standards.

Dental Intern

Oct 2022 – Oct 2023

Government Dental College and Hospital

Hyderabad, India

- Assessed oral health data from screenings and public health surveys to identify disease trends. Evaluated community dental programs by tracking patient outcomes and intervention success rates.

Projects

Cloud-Based Healthcare Data Analysis

Jan 2025 – May 2025

- Automated web scraping with Python/BeautifulSoup for 50K+ product listings from 5 sites, cutting manual collection time by 90%.
- Built a scalable Python framework using pandas, yfinance, and BeautifulSoup, capable of processing large-scale healthcare utilization and cost datasets across 10+ time intervals (1d to 10y).
- Achieved 100% process automation, creating a reusable codebase capable of handling 230M+ records without performance degradation, supporting healthcare cost and claims analysis workflows.

Skills: Python, Pandas, Web Scraping, Data Automation, Healthcare Cost Analysis, Big Data Processing, ETL Pipelines, HL7/FHIR Standards, SQL, Power BI, HIPAA Compliance

Airline Data Analysis & Predictive Modeling for Flight Delays

Jan 2025 – May 2025

- Analyzed 10,000+ flight records using EDA, ANOVA, and regression, uncovering that class type influenced prices by up to 42%, while advance booking and distance had negligible impact—insights that informed a 15% improvement in revenue optimization strategies.
- Designed and deployed predictive models (Linear Regression, Random Forest), yielding RMSE 561 and improving accuracy by 18%, demonstrating experience in risk prediction modeling methods used in healthcare analytics.
- Visualized pricing trends using Matplotlib and Seaborn, and delivered data-backed recommendations to stakeholders, enhancing customer segmentation accuracy by 25%.

- Applied machine learning workflows including feature engineering, cross-validation, and hyperparameter tuning, reducing model overfitting by 12% and boosting long-term generalizability for healthcare data applications.

Skills: Python, Scikit-learn, Regression Modeling, Machine Learning, EDA, Data Visualization, Predictive Analytics, Risk Modeling, SAS/SPSS, Healthcare Analytics Applications

Optimizing Healthcare Mergers: Data-Driven Insights for Mayo Clinic Oct 2024 – Dec 2024

- Conducted market intelligence analysis and developed a multi-channel outreach strategy projected to reach 500K+ patients across digital, social media, and provider networks aligning with merger objectives.
- Performed financial benchmarking and competitive pricing analysis using real-world hospital data, building a strategic pricing framework that increased new patient acquisition by 15% in Q1 and ensured alignment with value-based care reimbursement models.
- Designed a merger impact simulation integrating clinical outcomes, financial metrics, and patient engagement data, projecting a 12% improvement in care accessibility, a 10% reduction in operational costs, and a 9% increase in patient satisfaction post-merger.

Skills: Healthcare Analytics, Market Intelligence, Financial Benchmarking, Value-Based Care, Epic Systems, SWOT Analysis, Strategic Planning, Balanced Scorecard, Healthcare Trends Evaluation

Healthcare Acquisition Impact Analysis Aug 2024 – Nov 2024

- Conducted in-depth analysis of telehealth integration into managed care systems, revealing a 50% increase in utilization from pre-2020 to 2022, significantly expanding access to care during and after the COVID-19 pandemic.
- Evaluated patient-reported experience measures (PREMs) and patient-reported outcome measures (PROMs), showing 80%+ satisfaction rates, primarily driven by improved convenience, continuity of care, and reduced appointment wait times—critical metrics in value-based care models.
- Estimated claims data and care delivery patterns to quantify cost savings, projecting a 12% reduction in unnecessary emergency room visits and a 15% improvement in chronic disease management adherence through telehealth adoption.

Skills: Telehealth Analytics, Claims Data Analysis, Patient-Reported Outcomes, Value-Based Care, EHR Systems, Healthcare Informatics, Workflow Optimization, Six Sigma, Digital Health Policy

SKILLS

Programming & Analytics: Python, R, SQL, SAS, SPSS, Pandas, NumPy, Scikit-learn, Statistical Modeling, Predictive Analytics

Visualization Tools: Power BI, Tableau, Microsoft Excel, Matplotlib, Seaborn, Data Storytelling, Dashboard Development

Project Management: Agile, Waterfall, MS Project, MS Teams, Gantt Charts, SWOT Analysis, Balanced Scorecard, Strategic Planning

Healthcare Standards & Informatics: HL7, ICD-10, RxNorm, SNOMED CT, LOINC, CPT, HIPAA Compliance, EHR Systems (Epic, Cerner), Telehealth Integration, Value-Based Care Models

Other Professional Skills: Workflow Optimization, Six Sigma, Market Research, Competitive Analysis, Digital Health Policy, Data Interpretation, Research Methodology

Seminars and Conferences

- Attended 73rd Indian dental conference Jan 2020, Thiruvananthapuram
- Attended 7th Telangana state dental conference. March 2022
- Attended at Hyderabad Dental Show, July 2022.